

Menu-Driven C Program for Digit Analysis

Technical Documentation & Test Specifications

1. Algorithm

- Step 1:** Start.
- Step 2:** Declare variables: `number` (to store the user input), `choice` (for menu selection), `temp` (for temporary calculations), and an integer array `freq[10]` initialized to 0.
- Step 3:** Prompt the user to input a positive integer and read it into `number`.
- Step 4:** Display the menu options:
- 1. Check whether all its digits are unique.
 - 2. Find the frequency of each digit from 0 to 9.
- Step 5:** Read the user's `choice`.
- Step 6:** Assign `number` to `temp`. If `temp` is 0, set `freq[0] = 1`. Otherwise, use a loop (`while temp > 0`) to extract the last digit (`temp % 10`), increment the corresponding index in `freq` array, and remove the last digit (`temp = temp / 10`).
- Step 7:** Use a `switch` statement on `choice`:
- Case 1 (Unique Digits):** Iterate through the `freq` array. If any element `freq[i] > 1`, set a flag `isUnique = 0` and break. Print "Unique" if `isUnique` is 1, else print "Not Unique".
 - Case 2 (Digit Frequency):** Iterate through the `freq` array from index 0 to 9. Print the frequency of each index formatted as `freq_i = count`.
 - Default:** Print "Invalid choice!".
- Step 8:** Stop.

2. Test Case Table

Test Case ID	Input Number	Menu Choice	Expected Output	Result
TC_01	12345	1	Unique	Pass
TC_02	12234	1	Not Unique	Pass
TC_03	112333	2	freq_0 = 0, freq_1 = 2, freq_2 = 1, freq_3 = 3, freq_4 = 0, freq_5 = 0, freq_6 = 0, freq_7 = 0, freq_8 = 0, freq_9 = 0	Pass
TC_04	9	1	Unique	Pass
TC_05	77	9	Invalid choice!	Pass

```
(kali㉿kali)-[~/demo1]  
└─$ cc algo.c
```

```
(kali㉿kali)-[~/demo1]  
└─$ ./a.out
```

Enter a positive integer: 12345

— MENU —

1. Check whether all its digits are unique
2. Find the frequency of each digit from 0 to 9

Enter your choice (1 or 2): 1

Output: Unique

```
(kali㉿kali)-[~/demo1]  
└─$ ./a.out
```

Enter a positive integer:

12345

— MENU —

1. Check whether all its digits are unique
2. Find the frequency of each digit from 0 to 9

Enter your choice (1 or 2): 2

freq_0 = 0, freq_1 = 1, freq_2 = 1, freq_3 = 1, freq_4 = 1, freq_5 = 1, freq_6 = 0, freq_7 = 0, freq_8 = 0, freq_9 = 0

3. C Program Code

```
#include <stdio.h>

int main() {
    long long number, temp;
    int choice;
    int freq[10] = {0};
    int isUnique = 1;

    // Input the number
    printf("Enter a positive integer: ");
    scanf("%lld", &number);

    // Display Menu
    printf("\n--- MENU ---\n");
    printf("1. Check whether all its digits are unique\n");
    printf("2. Find the frequency of each digit from 0 to 9\n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    // Calculate digit frequencies
    temp = number;
    if (temp == 0) {
        freq[0] = 1;
    } else {
        while(temp > 0) {
            int digit = temp % 10;
            freq[digit]++;
            temp /= 10;
        }
    }

    // Switch case for menu driven operations
    switch(choice) {
        case 1:
            for(int i = 0; i < 10; i++) {
                if(freq[i] > 1) {
                    isUnique = 0;
                    break;
                }
            }
            if(isUnique)
                printf("Output: Unique\n");
            else
                printf("Output: Not Unique\n");
            break;

        case 2:
            for(int i = 0; i < 10; i++) {
                printf("freq_%d = %d", i, freq[i]);
                if(i < 9) printf(", ");
            }
            printf("\n");
            break;

        default:
            printf("Invalid choice! Please select 1 or 2.\n");
    }

    return 0;
}
```

4. Test Execution & Output (Simulated)

Run 1 (TC_01):

Enter a positive integer: 12345

--- MENU ---

1. Check whether all its digits are unique
2. Find the frequency of each digit from 0 to 9

Enter your choice (1 or 2): 1

Output: Unique

Run 2 (TC_02):

Enter a positive integer: 12234

--- MENU ---

1. Check whether all its digits are unique
2. Find the frequency of each digit from 0 to 9

Enter your choice (1 or 2): 1

Output: Not Unique

Run 3 (TC_03):

Enter a positive integer: 112333

--- MENU ---

1. Check whether all its digits are unique
2. Find the frequency of each digit from 0 to 9

Enter your choice (1 or 2): 2

freq_0 = 0, freq_1 = 2, freq_2 = 1, freq_3 = 3, freq_4 = 0, freq_5 = 0, freq_6 = 0, freq_7 = 0,
freq_8 = 0, freq_9 = 0